BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 073 · Lesson 073 · Trace Tables and Boundary Testing

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does CountPasses count?

Q2. What is a trace table for?

Q3. What is a boundary condition?

Q4. Stack Push 3, Push 7, then Pop. What is popped and what remains?

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I Do · Read new knowledge

Key vocabulary

trace table — State changes in algorithm execution.

sentinel — Special value indicating a condition.

boundary test — Values at or near limit.

exceptional test — Unusual or error-causing case.

New learning

Observe — A trace table records variable values after each meaningful execution step: i, a[i], condition and result.

Explain — A boundary test probes just below, at and above a threshold. Normal, erroneous and exceptional tests reveal different faults.

Evaluate — Test early success, late success, absent target, and empty arrays. For this function, −1 distinguishes not found from index0.

Worked example

Trace first even index in [3,5,8,10].

1. Explain — The function exits at first even occurrence and does not examine10.

We Do · Practise with guidance

Trace first even index in [1,7,9], including return when absent.

```
i0 1 odd; i1 7 odd; i2 9 odd; loop ends, returns−1.
```

Explain

No candidate satisfies value%2==0, so the not-found sentinel is returned.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace FindFirstEven([7,9,6,4]) with index and parity. [6]

Question 2

Identify expected outputs for [], [2,3], [3,5,8], [1,7,9]. [4]

Question 3

A program incorrectly returns0 when there is no even value. Explain the ambiguity and propose a safer fix, with tests. [6]

Exit Ticket · Answer before checking

Returning0 is a safe 'not found' value for zero-indexed arrays.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Array values greater than or equal to40.

Q2. Showing changing variables/decisions.

Q3. A limit at which program behaviour changes.

Q4. Popped 7; the stack contains 3.

You Do pages · Independent questions

Q1. i0 7 odd [1], i1 9 odd [1], i2 6 even [2], return index2 [1], do not inspect4 [1].

Q2. [] −1 [1]; [2,3] 0 [1]; [3,5,8] 2 [1]; [1,7,9] −1 [1].

Q3. 0 is valid first index [2]; absent must return distinct sentinel −1 [2]; test empty and first-item-even to distinguish [2].

Exit Ticket · Model answer

False. Index0 is a valid result.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: